

Section 3: Predicting Portfolio Risk

3.1–3.2: Return statistics

Table 1 shows the sample means and standard deviations of monthly excess returns for the three long-short factors (Value, Momentum, Quality) and the market over the sample period.

	Market	Value	Momentum	Quality
Mean	0.0069	0.0032	0.0033	0.0031
SD	0.0451	0.0393	0.0362	0.0164

3.3: Correlation matrix

Table 2 reports the correlation matrix of the same four return series.

	Market	Value	Momentum	Quality
Market	1.00	-0.27	-0.34	-0.18
Value	-0.27	1.00	-0.25	-0.14
Momentum	-0.34	-0.25	1.00	0.23
Quality	-0.18	-0.14	0.23	1.00

All three factors (Value, Momentum, Quality) are negatively correlated with the market, consistent with them being long-short “hedged” strategies. Value and Momentum are negatively correlated (−0.25), Value and Quality are negatively correlated (−0.14), and Momentum and Quality are positively correlated (0.23), offering less diversification benefit as a pair.

3.4: Portfolio risk

The Portfolio allocates 50% to Value, 25% to Momentum, and 25% to Quality. Using the general portfolio variance formula,

$$Var(P) = \sum_i \sum_j w_i w_j \sigma_i \sigma_j corr(i, j), \quad SD(P) = \sqrt{Var(P)},$$

with the sample standard deviations and correlations in 3.1-3.3, the predicted standard deviation of The Portfolio is 1.97% per month.